Risk-Based Inspection (RBI) Software

Study Summary, Functional Understanding & Proposed Development Approach

Prepared as an internal study document based on the RBI basics discussed with the engineer/client.

Purpose of This Document

I studied the basic concept and workflow of Risk-Based Inspection (RBI) software. This document summarizes my understanding of how an RBI system works, what information it needs, which software modules are required, how risk assessment connects to inspection planning, and how Artificial Intelligence can be added in the future. It also explains how we can approach the development of the requested software in phases.

> **Important:** Exact engineering formulas, scoring rules, inspection intervals and risk thresholds must be finalized according to the client's selected RBI methodology and validated by a qualified engineer. This document explains the software workflow and conceptual model, not a final engineering calculation standard.

1. My Understanding of RBI Software

The main purpose of RBI is to prioritize inspection based on the actual risk of equipment instead of relying only on fixed calendar-based inspection schedules. The core concept combines Probability of Failure (PoF) and Consequence of Failure (CoF) to determine equipment risk.

> **Overall workflow:** Plant → Equipment → Operating Data → Damage Mechanisms → PoF → CoF → Risk → Inspection Plan → Inspection Results → Reassessment

2. Plant and Equipment Structure

The software will create a digital structure of the plant. A plant can contain multiple units or areas, and each unit can contain different equipment.

Example: Plant → Process Unit → Pressure Vessel V-101

Equipment Data	Example
Equipment ID	V-101
Equipment Type	Pressure Vessel
Material	Carbon Steel
Design Pressure	20 bar
Operating Pressure	16 bar
Design Temperature	180°C
Operating Temperature	150°C
Installation Year	2012
Fluid	Hydrocarbon

3. Operating Conditions

The system needs operating information because risk depends on actual service conditions. Inputs may include pressure, temperature, flow, fluid composition, water content, chlorides, H■S, CO■, pH and operating cycles. In the future, these values can also be integrated from SCADA, PLCs, historians or other industrial systems.

4. Damage Mechanisms

The system needs to identify and record possible damage mechanisms for each equipment item. Examples include general corrosion, localized corrosion, pitting, erosion, cracking, fatigue, creep and hydrogen-related damage. Damage mechanism selection should be supported by engineering rules and reviewed by a qualified engineer. AI can assist with

recommendations but should not make uncontrolled engineering decisions.

5. Inspection History and Thickness Data

Previous inspection results are a major RBI input. UT thickness measurements should be stored with inspection date and measurement location. Historical data allows the system to identify thickness loss and trends.

6. Corrosion Rate and Remaining Life

For example, if thickness changes from 14.2 mm in 2022 to 11.9 mm in 2026, the loss is 2.3 mm and the simplified average corrosion rate is approximately 0.575 mm/year. If current thickness is 11.9 mm and minimum allowable thickness is 8 mm, the thickness margin is 3.9 mm. A simple numerical estimate gives about 6.8 years at that rate. In production, uncertainty, short/long-term rates and applicable engineering/code calculations must also be considered.

7. Probability of Failure (PoF)

PoF represents how likely an equipment item is to fail. Potential inputs include equipment age, corrosion/damage trends, thickness margin, damage mechanism, inspection history and operating conditions. An MVP can use configurable categorical scoring such as 1–5, while the final methodology must be defined by the client/engineer.

8. Consequence of Failure (CoF)

CoF represents the severity of the impact if equipment fails. Depending on the selected methodology, this can include personnel safety, fire/explosion potential, toxic exposure, environmental impact, production loss, repair cost and financial impact. For example, water service and flammable hydrocarbon service can have very different consequences.

9. Risk Calculation

Conceptually, Risk = Probability of Failure × Consequence of Failure. If PoF = 4 and CoF = 5, the numerical score is 20. The software can map the score into Low, Medium, High or Very High using a configurable risk matrix. The actual matrix must follow the client's approved methodology.

10. Risk Ranking and Inspection Planning

After risk assessment, the system will rank equipment across the plant. Higher-risk equipment receives higher inspection priority. The system can track inspection due dates, recommended methods, priorities, assignments and status so inspection resources can be focused where risk is highest.

11. Inspection Effectiveness

The system should record not only that an inspection occurred, but also the inspection method and relevant effectiveness/coverage information required by the selected RBI methodology. For example, visual inspection and UT provide different types of information.

12. Reassessment After Inspection

RBI should be a closed-loop system. New inspection measurements and findings are added to the database, applicable trends/calculations are reassessed, and PoF, risk and inspection planning can be updated. This makes the platform a living risk-management system rather than a one-time report generator.

13. Proposed Software Modules

1. User & Role Management
2. Plant, Unit and Area Management
3. Equipment Register
4. Materials and Process Data
5. Damage Mechanism Management
6. Inspection History and Measurements
7. PoF Assessment Engine
8. CoF Assessment Engine
9. Risk Calculation and Risk Matrix
10. Inspection Planning and Scheduling
11. Risk and Inspection Dashboard
12. PDF/Excel Reporting
13. Audit Trail and Approval Workflow

14. Proposed Database Structure

A PostgreSQL database can initially include: users, plants, units, equipment, equipment_types, materials, fluids, operating_conditions, damage_mechanisms, equipment_damage_mechanisms, inspection_records, inspection_measurements, inspection_methods, pof_assessments, cof_assessments, risk_assessments, risk_matrix, inspection_plans, recommendations, documents and audit_logs.

15. Proposed Technical Architecture

Frontend: React/Next.js web dashboard
Backend: FastAPI/Python
Database: PostgreSQL
Document Storage: Secure storage for inspection reports
Calculation Layer: Separate deterministic RBI/engineering calculation services
AI Layer: Separate AI services for extraction, analysis, explanation and assistant features
Integration Layer: APIs for ERP, SCADA/PLC, historian or other plant systems.

16. Future AI Integration

AI Document Extraction

Upload old inspection PDFs, scanned reports or Excel files. AI can extract equipment IDs, inspection dates, thickness values, locations and findings. Extracted information should be shown for engineer review before saving.

AI Trend Detection

Analyze historical thickness and inspection data to flag increasing corrosion trends, abnormal changes or unusual patterns.

AI Risk Explanation

Instead of only showing 'High Risk', explain which factors contributed to the risk classification.

AI Inspection Assistant

Allow engineers to ask questions such as: What is the current risk? What were the last inspection findings? Which equipment is due within 90 days?

What-if Analysis

Allow engineers to test scenarios such as how a higher corrosion rate could affect remaining life, risk and inspection priority.

AI Report Generation

Generate draft management summaries and reports from approved assessment data.

Condition Monitoring Integration

In the future, connect pressure, temperature, flow, vibration and other live industrial data so significant operating changes can trigger reassessment or engineering review.

17. Proposed AI Architecture

A future AI layer can use an orchestrator to coordinate specialized services:

> **RBI AI Orchestrator** → **Document Agent** → **Equipment/Data Agent** → **Damage/Knowledge Assistant** → **PoF/CoF Calculation Services** → **Risk Agent** → **Inspection/Recommendation Agent**

Key principle: AI should not replace deterministic engineering calculations. Approved formulas, scoring rules and methodology should remain in a version-controlled calculation engine. AI should primarily assist with document extraction, knowledge retrieval, explanations, trend/anomaly flagging, natural-language interaction and recommendations. Final engineering approval should remain with a qualified engineer.

18. Recommended MVP Development Plan

Phase 1 — Foundation

User management, plant/unit/equipment register, PostgreSQL database, equipment data and inspection history.

Phase 2 — RBI Core

Damage mechanisms, configurable PoF/CoF inputs, risk matrix, risk calculation and ranking.

Phase 3 — Inspection Management

Inspection planning, due dates, methods, findings, measurement history and reassessment workflow.

Phase 4 — Dashboard & Reports

Risk dashboard, equipment drill-down, inspection schedule, trends and PDF/Excel reports.

Phase 5 — AI Layer

PDF/Excel extraction, natural-language RBI assistant, risk explanations, trend/anomaly flags and report drafting.

Phase 6 — Industrial Integration

Optional SCADA/PLC/historian/ERP integrations and condition-based reassessment triggers.

19. Recommended MVP Scope

I recommend starting with Pressure Vessels and Piping rather than trying to build a complete refinery-scale platform immediately. The MVP can include equipment register, operating data, inspection history, thickness measurements, corrosion trend, damage mechanism selection, configurable PoF/CoF assessment, risk matrix, risk ranking, inspection planning, dashboard and reports. Tanks, heat exchangers, boilers and more advanced methodologies can be added later.

20. Standards and Engineering Validation

Commonly relevant references may include API RP 580 (RBI framework), API RP 581 (RBI methodology), API 510 (pressure vessel inspection), API 570 (piping inspection) and API 653 (tank inspection). The actual implementation must

follow the client's adopted standard/edition, applicable jurisdiction, licensing/copyright requirements and qualified engineering review.

21. My Final Understanding

After studying the RBI concept, I understand that RBI software is not simply a dashboard or risk matrix. It is a complete workflow that starts with plant/equipment data, operating conditions, damage mechanisms and inspection history; evaluates PoF and CoF; determines risk; creates an inspection priority/plan; and uses new inspection results to reassess the equipment. Therefore, the system should be designed as a modular and continuously updated risk-management platform.

From the software-development perspective, I believe we can build this system in stages. We can first develop the deterministic RBI core and inspection management system, then add the AI layer. The strongest future AI use cases are inspection-report data extraction, historical trend analysis, risk explanation, natural-language assistance, what-if analysis, anomaly flagging and management reporting. Later, live industrial data can be integrated to move toward a condition-aware Industry 4.0 RBI solution.

22. Information to Confirm with the Engineer Before Development

1. Exact RBI standard/methodology and edition
2. Equipment types required in the first version
3. Exact PoF and CoF calculation/scoring methodology
4. Approved risk matrix and category thresholds
5. Initial damage mechanism list
6. Available Excel/database/inspection reports
7. Historical thickness/inspection data format
8. Inspection methods and effectiveness/coverage rules
9. Engineering logic for inspection intervals
10. Assessment review and approval process
11. Required report formats
12. Future ERP/SCADA/PLC/historian integration requirements

Conclusion

Overall, I understand the project as technically feasible from a software-development perspective. The most important first step is to define and validate the exact RBI calculation methodology with the engineer/client. Once that is clear, we can build the system as modular components: equipment and inspection management, engineering calculation engine, risk engine, inspection planning, dashboards, reporting and finally an AI layer. This approach also keeps the platform flexible for future industrial integrations and advanced AI capabilities.